ASSIGNMENT # 6

View:

1. Create a table actor1(drop if already exists) from actor having three columns actor_id, name (concatenate first_name & last_name) for first 10 actors.

```
mysql> use sakila;
Database changed
```

```
mysql> create table actor1
    -> (
    -> actor_id smallint(5) primary key,
    -> name varchar(90),
    -> last_update timestamp
    -> );
Query OK, 0 rows affected (0.17 sec)
```

```
mysql> insert into actor1
    -> (select actor_id, concat(first_name," ",last_name), last_update from actor limit 10);
Query OK, 10 rows affected (0.03 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> select * from actor1;
+----------+---------------------+---------------------+
| actor_id | name                | last_update         |
+----------+---------------------+---------------------+
|        1 | PENELOPE GUINESS    | 2006-02-15 18:04:33 |
|        2 | NICK WAHLBERG       | 2006-02-15 18:04:33 |
|        3 | ED CHASE            | 2006-02-15 18:04:33 |
|        4 | JENNIFER DAVIS      | 2006-02-15 18:04:33 |
|        5 | JOHNNY LOLLOBRIGIDA | 2006-02-15 18:04:33 |
|        6 | BETTE NICHOLSON     | 2006-02-15 18:04:33 |
|        7 | GRACE MOSTEL        | 2006-02-15 18:04:33 |
|        8 | MATTHEW JOHANSSON   | 2006-02-15 18:04:33 |
|        9 | JOE SWANK           | 2006-02-15 18:04:33 |
|       10 | CHRISTIAN GABLE     | 2006-02-15 18:04:33 |
+----------+---------------------+---------------------+
10 rows in set (0.00 sec)
```

2. Create a View view1 having two columns actor_id & name Display View1.

```
mysql> create view view1 as
    -> select actor_id, name
    -> from actor1;
Query OK, 0 rows affected (0.03 sec)

mysql> select * from view1;
+----------+---------------------+
| actor_id | name                |
+----------+---------------------+
|        1 | PENELOPE GUINESS    |
|        2 | NICK WAHLBERG       |
|        3 | ED CHASE            |
|        4 | JENNIFER DAVIS      |
|        5 | JOHNNY LOLLOBRIGIDA |
|        6 | BETTE NICHOLSON     |
|        7 | GRACE MOSTEL        |
|        8 | MATTHEW JOHANSSON   |
|        9 | JOE SWANK           |
|       10 | CHRISTIAN GABLE     |
+----------+---------------------+
10 rows in set (0.00 sec)
```

3. Add one more column last_updated to view1.

```
mysql> alter view view1
    -> as
    -> select actor_id,name,last_update from actor1;
Query OK, 0 rows affected (0.01 sec)
```

4. Display View1 & Drop view1.

```
mysql> select * from view1;
+----------+---------------------+---------------------+
| actor_id | name                | last_update         |
+----------+---------------------+---------------------+
|        1 | PENELOPE GUINESS    | 2006-02-15 18:04:33 |
|        2 | NICK WAHLBERG       | 2006-02-15 18:04:33 |
|        3 | ED CHASE            | 2006-02-15 18:04:33 |
|        4 | JENNIFER DAVIS      | 2006-02-15 18:04:33 |
|        5 | JOHNNY LOLLOBRIGIDA | 2006-02-15 18:04:33 |
|        6 | BETTE NICHOLSON     | 2006-02-15 18:04:33 |
|        7 | GRACE MOSTEL        | 2006-02-15 18:04:33 |
|        8 | MATTHEW JOHANSSON   | 2006-02-15 18:04:33 |
|        9 | JOE SWANK           | 2006-02-15 18:04:33 |
|       10 | CHRISTIAN GABLE     | 2006-02-15 18:04:33 |
+----------+---------------------+---------------------+
10 rows in set (0.00 sec)

mysql> drop view view1;
Query OK, 0 rows affected (0.00 sec)
```

Index:

1. Create an index – name_Index for table actor1 on filed Name (display only name of actor's before creating the index and after creating the index).

```
mysql> select name from actor1;
+---------------------+
| name                |
+---------------------+
| PENELOPE GUINESS    |
| NICK WAHLBERG       |
| ED CHASE            |
| JENNIFER DAVIS      |
| JOHNNY LOLLOBRIGIDA |
| BETTE NICHOLSON     |
| GRACE MOSTEL        |
| MATTHEW JOHANSSON   |
| JOE SWANK           |
| CHRISTIAN GABLE     |
+---------------------+
10 rows in set (0.00 sec)
```

```
mysql> create index name_index
    -> on actor1
    -> (name);
Query OK, 0 rows affected (0.21 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select name from actor1;
+---------------------+
| name                |
+---------------------+
| BETTE NICHOLSON     |
| CHRISTIAN GABLE     |
| ED CHASE            |
| GRACE MOSTEL        |
| JENNIFER DAVIS      |
| JOE SWANK           |
| JOHNNY LOLLOBRIGIDA |
| MATTHEW JOHANSSON   |
| NICK WAHLBERG       |
| PENELOPE GUINESS    |
+---------------------+
10 rows in set (0.00 sec)
```

2. Display the definition of First_Index

```
mysql> show indexes from actor1;
+--------+------------+------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+
| Table  | Non_unique | Key_name   | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment |
+--------+------------+------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+
| actor1 |          0 | PRIMARY    |            1 | actor_id    | A         |          10 |     NULL | NULL   |      | BTREE      |         |               |
| actor1 |          1 | name_index |            1 | name        | A         |          10 |     NULL | NULL   | YES  | BTREE      |         |               |
+--------+------------+------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+
2 rows in set (0.00 sec)
```

3. Drop name_index.

```
mysql> drop index name_index on actor1;
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Function (user define or stored function):

Drop last_update & add a new column INCOME(decimal(10,2)) to actor1 and fill the values:

Now create a function to check the income level of actor's as per given below conditions:

- If income < 1000 then income level is "Low Income"
- If income >1000 and income <2000 then income level is "Avg Income"
- If income > 2000 then income level is "High Income".

Use delimiter(//,$$, etc..) to execute multiple lines at the same time

```
mysql> alter table actor1
    -> drop last_update;
    -> //
Query OK, 0 rows affected (0.38 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> alter table actor1
    -> add income decimal(10,2);
Query OK, 0 rows affected (0.33 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> update actor1
    -> set income = 1100 where actor_id=2;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update actor1
    -> set income = 900 where actor_id=3;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update actor1
    -> set income = 650 where actor_id=4;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update actor1
    -> set income = 1350 where actor_id=5;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update actor1
    -> set income = 2000 where actor_id=6;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update actor1
    -> set income = 150 where actor_id=7;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update actor1
    -> set income = 1650 where actor_id=8;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update actor1
    -> set income = 2100 where actor_id=9;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update actor1
    -> set income = 2000 where actor_id=10;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from actor1;
    -> //
+----------+---------------------+---------+
| actor_id | name                | income  |
+----------+---------------------+---------+
|        1 | PENELOPE GUINESS    | 1000.00 |
|        2 | NICK WAHLBERG       | 1100.00 |
|        3 | ED CHASE            |  900.00 |
|        4 | JENNIFER DAVIS      |  650.00 |
|        5 | JOHNNY LOLLOBRIGIDA | 1350.00 |
|        6 | BETTE NICHOLSON     | 2000.00 |
|        7 | GRACE MOSTEL        |  150.00 |
|        8 | MATTHEW JOHANSSON   | 1650.00 |
|        9 | JOE SWANK           | 2100.00 |
|       10 | CHRISTIAN GABLE     | 2000.00 |
+----------+---------------------+---------+
10 rows in set (0.00 sec)
```

```
mysql> create function cal(income decimal(10,2) returns varchar(30)
    -> begin
    -> declare income_level varchar(50);
    -> if income<1000 then
    -> set income_level="Low Income";
    -> elseif income>1000 and income<2000 then
    -> set income_level="Avg Income";
    -> else
    -> set income_level="High Income";
    -> end if;
    -> return income_level;
    -> end
    -> //
```

```
mysql> select actor_id,name,cal(income) from actor1;
    -> //
+----------+---------------------+-------------+
| actor_id | name                | cal(income) |
+----------+---------------------+-------------+
|        1 | PENELOPE GUINESS    | High Income |
|        2 | NICK WAHLBERG       | Avg Income  |
|        3 | ED CHASE            | Low Income  |
|        4 | JENNIFER DAVIS      | Low Income  |
|        5 | JOHNNY LOLLOBRIGIDA | Avg Income  |
|        6 | BETTE NICHOLSON     | High Income |
|        7 | GRACE MOSTEL        | Low Income  |
|        8 | MATTHEW JOHANSSON   | Avg Income  |
|        9 | JOE SWANK           | High Income |
|       10 | CHRISTIAN GABLE     | High Income |
+----------+---------------------+-------------+
10 rows in set (0.00 sec)
```

```
mysql> drop table actor1;
Query OK, 0 rows affected (0.08 sec)
```